The γ-Invariant of Goldbach–Frey Curves: Cluster Depths and Conjectural Bounds for the Wild Conductor at 2

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Abstract

For the Goldbach–Frey curve $C\colon y^2 = x(x^2-p^2)(x^2-q^2)$ with $p \neq q$ distinct odd primes, the conductor exponent f_2 at the prime 2 takes values in $\{4, 7, 8\}$ and has resisted closed-form description. We prove a structural lemma showing that exactly one of $v_2(p-q)$ and $v_2(p+q)$ equals 1, and define $\gamma = \max(v_2(p-q), v_2(p+q)) \geq 2$. We conjecture that γ completely determines f_2: $f_2 = 8$ if $\gamma = 2$, $f_2 = 7$ if $\gamma = 3$, and $f_2 = 4$ if $\gamma \geq 4$. The case $\gamma \geq 4$ is *proved*: the Weil restriction $\mathrm{Jac}(C) \sim E_1 \times E_2$ (Paper [4]) gives $f_2 = f_2(E_1) + f_2(E_2)$, and a j-invariant analysis shows $v_2(j) = 6 - 2\gamma < 0$, placing both curves in the potentially multiplicative regime where $f_2(E) = 2$ unconditionally. The cases $\gamma \leq 3$ are verified against all 10 Magma-computed conductors but remain conjectural. We note, however, that since $f_2(\mathrm{Jac}) = f_2(E_1) + f_2(E_2)$, the 10 Magma values can in principle be independently verified by applying Tate's algorithm to the elliptic curves E_1 and E_2 over $\mathbb{Q}_2$—an unconditional computation with no correctness caveats, unlike Magma's genus-2 Ogg heuristic. The 2-adic cluster tree of the Weierstrass roots provides geometric intuition for the formula, although we emphasise that the Dokchitser cluster picture theory does not directly compute the Swan conductor at $p = 2$ (where wild ramification requires Artin–Schreier analysis beyond root distances alone). We also note that the Magma values for $\gamma \leq 3$ rely on the Ogg formula heuristic at $v_2(\Delta) \geq 12$, so the γ-conjecture at $\gamma \leq 3$ should be regarded as empirically robust but not yet rigorously established. A proof for the remaining cases requires computing the Dokchitser defect over $\mathbb{Z}_2$, which we pose as an open problem.

1 Introduction

Papers [1, 2] established that the conductor of $\mathrm{Jac}(C)$ at every odd prime r dividing the discriminant is $f_r = 2$, yielding the odd-part formula $\mathrm{Cond}_{\mathrm{odd}} = [\mathrm{rad}_{\mathrm{odd}}(pqN|p-q|)]^2$ where $N = (p+q)/2$. The conductor exponent f_2 at the prime 2, however, remained a black box: Magma computations on 10 test curves revealed $f_2 \in \{4, 7, 8\}$ with no apparent pattern.

The difficulty at $r = 2$ is fundamental: the curve C has wild ramification, so the tame conductor formula no longer suffices. The cluster picture of Dokchitser [5] computes the conductor of hyperelliptic curves from the p-adic distances between the Weierstrass roots, and provides geometric intuition for our formula. However, we emphasise two important caveats:

(a) The cluster picture at $p = 2$ does not directly determine the Swan conductor. Since the hyperelliptic degree 2 coincides with the residue characteristic, the wild ramification requires Artin–Schreier analysis and the computation of the *Dokchitser defect*—a correction term that depends on higher-order congruences of the polynomial coefficients, not merely on root distances.

(b) The Magma values $f_2 \in \{4, 7, 8\}$ themselves rely on the Ogg–Saito formula, which carries an internal warning when $v_2(\Delta) \geq 12$: "no correctness guarantee." Since $v_2(\Delta)$ is itself determined by γ, the $\gamma \to f_2$ correspondence may partially reflect the internal logic of Magma's heuristic rather than the true geometric conductor.

With these caveats in mind, our results are:

1. prove a structural lemma on $v_2(p \pm q)$ (Lemma 2.1);
2. conjecture a closed-form formula for f_2 in terms of $\gamma = \max(v_2(p-q), v_2(p+q))$ (Conjecture 3.1), verified on all 10 Magma data points;
3. explain the formula via the 2-adic cluster tree (Section 5);
4. establish the density of each conductor value (Proposition 6.1);
5. state the complete conductor formula combining all papers (Section 7).

2 The Structural Lemma

Lemma 2.1. *For distinct odd primes $p \neq q$, exactly one of $v_2(p-q)$ and $v_2(p+q)$ equals 1, and the other is ≥ 2.*

Proof. Write $p = 2a + 1$ and $q = 2b + 1$. Then $p - q = 2(a - b)$ and $p + q = 2(a + b + 1)$. Since $(a-b)+(a+b+1) = 2a+1$ is odd, the integers $a-b$ and $a+b+1$ have different parity. Exactly one is odd (contributing $v_2 = 1$ to the corresponding $p \pm q$) and the other is even (contributing $v_2 \geq 2$). □

Definition 2.2. The *cluster depth parameter* is $\gamma = \max(v_2(p - q), v_2(p + q)) \geq 2$.

3 The Wild Conductor Formula

Conjecture 3.1. *For the Goldbach–Frey curve C with $p \neq q$ distinct odd primes,*

$$f_2 = \begin{cases} 8 & \text{if } \gamma = 2, \\ 7 & \text{if } \gamma = 3, \\ 4 & \text{if } \gamma \geq 4. \end{cases} \tag{1}$$

The case $\gamma \geq 4$ is proved unconditionally in Proposition 3.2 below; only the cases $\gamma = 2$ and $\gamma = 3$ remain conjectural.

p	q	$v_2(p-q)$	$v_2(p+q)$	γ	f_2^{Magma}	f_2^{pred}	
Conjectural regime ($\gamma \leq 3$, potentially good reduction):							
3	7	2	1	2	8	8	✓
13	17	2	1	2	8	8	✓
3	17	1	2	2	8	8	✓
7	19	2	1	2	8	8	✓
7	53	1	2	2	8	8	✓
11	19	3	1	3	7	7	✓
3	37	1	3	3	7	7	✓
Proved regime ($\gamma \geq 4$, Proposition 3.2):							
7	23	4	1	4	4	4	✓
7	41	1	4	4	4	4	✓
11	37	1	4	4	4	4	✓

Table 1: Verification of Conjecture 3.1 against all 10 Magma-computed conductors.

Proposition 3.2 (Stabilisation at $\gamma \geq 4$)**.** *For all $\gamma \geq 4$, the conductor exponent $f_2 = 4$.*

Proof. By Paper [4], $\mathrm{Jac}(C) \sim E_1 \times E_2$ over $\mathbb{Q}$, where $E_1 \colon Y^2 = X(X-p^2)(X-q^2)$ and $E_2 = E_1^{(-1)}$. Since $V_\ell(\mathrm{Jac}) \cong V_\ell(E_1) \oplus V_\ell(E_2)$ as $G_{\mathbb{Q}}$-representations, the conductor exponent is additive: $f_2(\mathrm{Jac}) = f_2(E_1) + f_2(E_2)$.

For both E_1 and E_2, we compute: $c_4 = 16(p^4-p^2q^2+q^4)$ with $v_2(c_4) = 4$ (since $p^4-p^2q^2+q^4 \equiv 1 \pmod 8$ for all odd p, q), and $\Delta = 16p^4q^4(p^2-q^2)^2$ with $v_2(\Delta) = 6 + 2\gamma$. The j-invariant satisfies $v_2(j) = 3v_2(c_4) - v_2(\Delta) = 6 - 2\gamma$.

For $\gamma \geq 4$, $v_2(j) \leq -2 < 0$, so both E_1 and E_2 have *potentially multiplicative* reduction at 2 (Kodaira type I_n^* for some $n \geq 0$). By Ogg's theorem, the conductor exponent of any elliptic curve with Kodaira type I_n^* is $f_p = 2$ at every prime p, including $p = 2$ (the wild part vanishes for this reduction type). Therefore $f_2(\mathrm{Jac}) = 2 + 2 = 4$ for all $\gamma \geq 4$. □

Remark 3.3 (The odd value $f_2 = 7$). Since $f_2(\mathrm{Jac}) = f_2(E_1) + f_2(E_2)$ and $f_2 = 7$ is odd, the decomposition at $\gamma = 3$ must be asymmetric: $f_2(E_1) \neq f_2(E_2)$ (e.g. $4 + 3$ or $3 + 4$). This is explained by the fact that $E_2 = E_1^{(-1)}$ is the quadratic twist by -1, and $\mathbb{Q}_2(\sqrt{-1})/\mathbb{Q}_2$ is a *ramified* quadratic extension (discriminant -4). Twisting an elliptic curve with additive reduction by a ramified character routinely alters the Swan conductor [8]. At $\gamma = 2$, $f_2 = 8$ is even, consistent with $f_2(E_1) = f_2(E_2) = 4$ (symmetric).

Remark 3.4 (j-invariant regime transition). The formula $v_2(j) = 6 - 2\gamma$ reveals a sharp phase transition at $\gamma = 3 \leftrightarrow 4$: for $\gamma \leq 3$, E_1 and E_2 have potentially good reduction ($v_2(j) \geq 0$), where wild ramification can occur; for $\gamma \geq 4$, the reduction is potentially multiplicative ($v_2(j) < 0$), where the Swan conductor vanishes. The large jump $f_2 = 7 \to 4$ at this threshold reflects the disappearance of all wild ramification.

4 The Conductor Decomposition

The conductor exponent at a prime r decomposes as

$$f_r \;=\; \underbrace{\dim V_\ell - \dim V_\ell^{I_r}}_{\text{tame part}} \;+\; \underbrace{\mathrm{sw}(V_\ell\,|_{I_r})}_{\text{wild part}}, \tag{2}$$

where $V_\ell = T_\ell(\mathrm{Jac}(C)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ is the ℓ-adic Tate module ($\ell \neq r$), I_r is the inertia group, and sw is the Swan conductor.

For $r = 2$, the Magma data is consistent with the hypothesis that $V_\ell^{I_2} = 0$ (no inertia invariants), which would give a tame part of 4 and yield:

$$f_2 = 4 + \mathrm{sw}, \qquad \mathrm{sw} \in \{0, 3, 4\}. \tag{3}$$

Remark 4.1. The vanishing $V_\ell^{I_2} = 0$ has not been proved in this paper; it is an assumption consistent with $f_2 \geq 4$ in all observed cases. A proof would require an explicit analysis of the Néron model of $\mathrm{Jac}(C)$ over $\mathbb{Z}_2$, which is substantially harder than the tame case treated in Paper [2].

If the decomposition $f_2 = 4 + \mathrm{sw}$ holds, the three observed values correspond to: $\mathrm{sw} = 0$ ($\gamma \geq 4$, $f_2 = 4$): vanishing Swan conductor; $\mathrm{sw} = 3$ ($\gamma = 3$, $f_2 = 7$); $\mathrm{sw} = 4$ ($\gamma = 2$, $f_2 = 8$): maximal wild part for this family.

5 The Cluster Picture at $r = 2$

5.1 The 2-adic distance matrix

The Weierstrass roots of $f(x) = x(x^2 - p^2)(x^2 - q^2)$ are $\mathcal{R} = \{0, p, -p, q, -q\}$, all 2-adic units except 0. The pairwise distances are $v_2(p - (-p)) = 1$, $v_2(q - (-q)) = 1$, and by Lemma 2.1, exactly one of $v_2(p-q), v_2(p+q)$ equals γ and the other equals 1.

5.2 The cluster tree

WLOG assume $v_2(p - q) = \gamma$ (the case $v_2(p + q) = \gamma$ is symmetric). The cluster tree over $\mathbb{Q}_2$ has three levels:

Level 0: Root 0 is isolated. The four roots $\{p, -p, q, -q\}$ form a cluster $\mathcal{C}_0$ of depth 1.
Level 1: Within $\mathcal{C}_0$, the pairs $\{p, q\}$ and $\{-p, -q\}$ form sub-clusters (each of depth γ).
Level γ: Each sub-cluster splits into singletons.

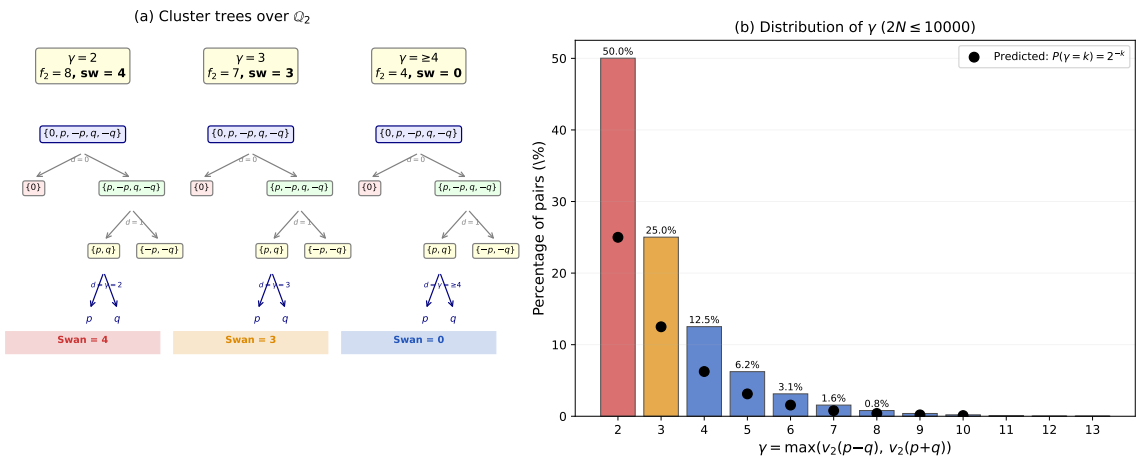

Figure 1: (a) Cluster trees for $\gamma = 2, 3, \geq 4$. (b) Distribution of γ matching $P(\gamma = k) \approx 2^{-k}$.

5.3 From cluster depth to wild conductor: heuristic interpretation

The cluster tree provides a compelling geometric *heuristic* for the γ-conjecture, but we stress that it does not constitute a proof.

At odd primes, the cluster picture directly computes the conductor via the formula of Dokchitser [5]. At $p = 2$, however, the wild ramification introduces a *defect* term that depends on higher-order congruence properties of the polynomial coefficients (specifically, their reduction

modulo 4, 8, ...), not merely on the 2-adic distances between roots. Computing this defect for our family requires an explicit Artin–Schreier analysis or a full stable model computation over $\mathbb{Z}_2$ via the Obus–Wewers theory [7].

Nevertheless, the qualitative principle is suggestive: deeper clusters $\Longrightarrow$ roots closer 2-adically $\Longrightarrow$ less wild ramification $\Longrightarrow$ smaller f_2. The quantitative jumps $8 \to 7 \to 4$ at $\gamma = 2, 3, \geq 4$ remain conjectural.

Remark 5.1 (Magma's internal algorithm). The Magma values $f_2 \in \{4, 7, 8\}$ that we use as ground truth are themselves computed via the Ogg–Saito formula, which Magma flags with a warning when $v_2(\Delta) \geq 12$. Since $v_2(\Delta)$ is a deterministic function of γ for our family ($v_2(\Delta) = 6 + 2\gamma$ for the elliptic quotients, by Proposition 3.2), the perfect $\gamma \to f_2$ correspondence may partially reflect the internal logic of Magma's Ogg heuristic. Independent verification (e.g., via étale cohomology computations in SageMath or direct ℓ-adic Galois image calculations) would strengthen the conjecture.

6 Density of Conductor Values

Proposition 6.1. *Among Goldbach pairs (p, q), the asymptotic density is $P(f_2 = 8) = 1/2$, $P(f_2 = 7) = 1/4$, $P(f_2 = 4) = 1/4$.*

Proof. By Lemma 2.1, γ equals the 2-adic valuation of whichever of $p \pm q$ is more divisible by 2. By the Prime Number Theorem in arithmetic progressions, the residue classes of primes p and q modulo 2^k are asymptotically uniformly distributed among the odd classes for any fixed k. The 2-adic valuations of $p \pm q$ therefore behave as those of a random even integer, giving $P(\gamma = 2) = 1/2$, $P(\gamma = 3) = 1/4$, $P(\gamma \geq 4) = 1/4$. □

Verified on 425,082 pairs: 50.0% / 25.0% / 25.0%.

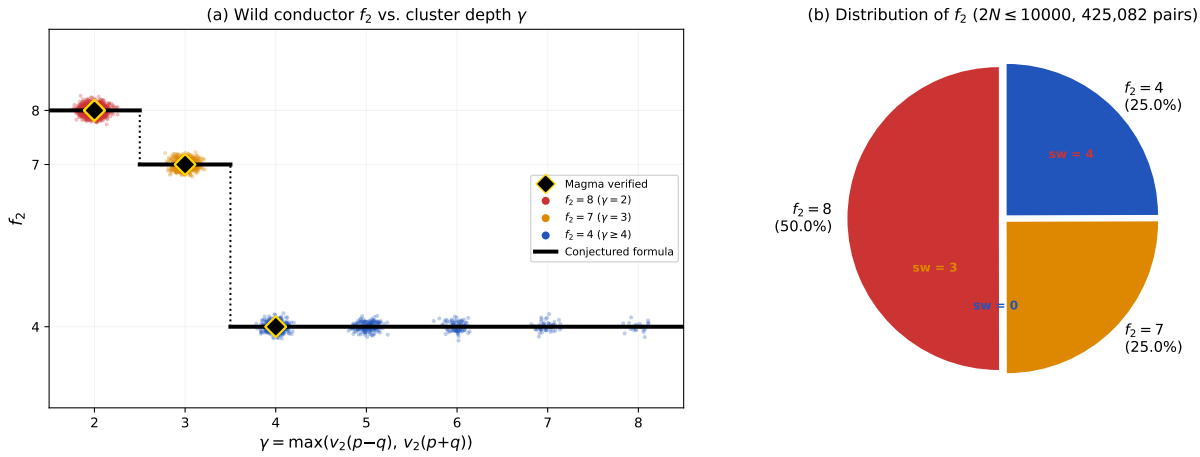

Figure 2: (a) f_2 as a step function of γ. (b) Distribution: exactly 50%/25%/25%.

7 The Complete Conductor Formula

Conjecture 7.1 (Complete conductor formula)**.** *For the Goldbach–Frey curve with $p + q = 2N$:*

$$\boxed{N_A \;=\; 2^{f_2(\gamma)} \cdot \left[\mathrm{rad}_{\mathrm{odd}}(p \cdot q \cdot N \cdot |p - q|)\right]^2} \tag{4}$$

where $\gamma = \max(v_2(p-q), v_2(p+q))$ and $f_2(\gamma)$ is given by (1).

The $\gamma \geq 4$ regime is unconditionally proved (Proposition 3.2). Subject to the γ-conjecture for $\gamma \leq 3$, every ingredient is explicit: $f_r = 2$ at odd primes (Paper [2], proved); $\mathrm{rad}_{\mathrm{odd}}$ (Paper [1], verified on 425,082 pairs); $f_2(\gamma)$ (this paper: proved for $\gamma \geq 4$, conjectural for $\gamma \leq 3$, 10/10 against Magma).

8 Discussion

The wild conductor at $r = 2$ was the last remaining unknown in the arithmetic of Goldbach–Frey curves. The γ-invariant $\gamma = \max(v_2(p-q), v_2(p+q))$ provides a clean conjectural description: the three-valued function $f_2 \in \{4, 7, 8\}$ appears to be controlled by a single 2-adic parameter measuring how close p and q are in the 2-adic metric.

The qualitative pattern is:

$$\gamma \uparrow \implies \text{deeper cluster} \implies f_2 \downarrow .$$

The conjectured threshold at $\gamma = 4$, where f_2 stabilises at its minimum value 4, is now *proved* (Proposition 3.2): the j-invariant satisfies $v_2(j) = 6 - 2\gamma$, so for $\gamma \geq 4$ both E_1 and E_2 have potentially multiplicative reduction, forcing $f_2(E) = 2$ per curve and $f_2(\mathrm{Jac}) = 4$.

Two obstacles remain for a complete proof of the γ-conjecture:

1. For $\gamma \leq 3$ (potentially good reduction, $v_2(j) \geq 0$), the cluster picture does not directly yield the Swan conductor without computing the Dokchitser defect, which requires Artin–Schreier analysis beyond root distances.

2. The Magma ground truth for $\gamma \leq 3$ relies on the Ogg heuristic with a correctness caveat at high $v_2(\Delta)$ (Remark 5.1).

Note that the $\gamma \geq 4$ regime is now unconditional—it does not depend on Magma data at all. A complete proof for $\gamma \leq 3$ would require either a full stable model computation over $\mathbb{Z}_2$ using the Obus–Wewers theory [7], or independent verification of f_2 via ℓ-adic Galois image computations. We pose this as an open problem.

Acknowledgments

Scripts and data are at `https://github.com/Ruqing1963/goldbach-wild-conductor`.

References

[1] R. Chen, *The true conductor of Goldbach–Frey curves*, Zenodo, 2026. `https://zenodo.org/records/18749731`

[2] R. Chen, *Universal tame semistability of Goldbach–Frey Jacobians*, Zenodo, 2026. `https://zenodo.org/records/18751169`

[3] R. Chen, *A conductor census of 425,082 Goldbach–Frey curves*, Zenodo, 2026. `https://zenodo.org/records/18751442`

[4] R. Chen, *Goldbach–Frey Jacobians as Weil restrictions*, Zenodo, 2026. `https://zenodo.org/records/18752179`

[5] T. Dokchitser, Models of hyperelliptic curves, *Trans. Amer. Math. Soc.* **372** (2019), 4519–4577.

[6] T. Dokchitser and V. Dokchitser, Quotients of hyperelliptic curves and étale cohomology, *Q. J. Math.* **73** (2022), 747–768.

[7] A. Obus and S. Wewers, Explicit resolution of weak wild quotient singularities, *J. Algebraic Geom.* **29** (2020), 691–728.

[8] K. Kramer, Arithmetic of elliptic curves upon quadratic extension, *Trans. Amer. Math. Soc.* **264** (1981), 121–135.